Reducing Liver Cancer Burden Through Hepatitis C Treatment Among People Who Have Injected Drugs in Singapore: A Modelling Study

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1 Assumptions

Population Structure and Incarceration

- The PWID population is stratified into nine age subgroups ($i = 1, \ldots, 9$) and four incarceration strata: never incarcerated (D), currently incarcerated (J), ever-incarcerated (F), and former PWID (X). Incarceration status affects transmission dynamics and treatment access but not disease progression rates.
- First arrest occurs at rate $\lambda_i^{(1)}$, release at rate $\lambda_i^{(2)}$, and re-arrest at rate $\lambda_i^{(3)}$, all varying by age subgroup i and fitted to Singapore Prison Service (SPS) data.
- Upon release, a proportion π of individuals return to active injecting and transition to the ever-incarcerated (F) stratum, while the remaining $(1 - \pi)$ cease injecting and transition to former PWID (X). Former PWID (X) are no longer at risk of HCV transmission or re-arrest.
- Entry into the PWID population occurs only into the never-incarcerated susceptible compartment $D_{u,1,i}$ at a time- and age-varying rate $\beta_i(t)$.

1.1 Transmission and Immunity

- HCV transmission occurs exclusively through needle-sharing among active PWID (strata D and F) at a per-event transmission probability q. Incarcerated individuals (J) and former PWID (X) are assumed to have no needle-sharing exposure.
- No sterilising immunity is assumed following either spontaneous clearance or treatment-induced SVR, consistent with the lack of consensus on protective immunity in the literature. Individuals who clear HCV — whether spontaneously or via DAA therapy — return to the susceptible state and remain at risk of reinfection upon re-exposure [12].
- Reinfection is defined as reappearance of HCV RNA after confirmed SVR in the presence of continued risk behaviour, consistent with EASL criteria requiring demonstration of a distinct genotype or phylogenetically distant strain from the primary infection [12]. Reinfection risk is therefore modelled identically to primary infection, governed by the same force of infection $\gamma_{i,j}(t)$.

1.2 Acute Infection and Spontaneous Clearance

- Acute HCV infection is defined as the period of up to 180 days (ι_1) following exposure, consistent with the clinical definition of no detectable anti-HCV antibody or negative HCV RNA within six months of presumed exposure [3].
- A proportion κ of acutely infected individuals achieve spontaneous clearance and return to the susceptible state. The remaining proportion $(1-\kappa)$ progress to chronic infection [9].
- Individuals undergoing DAA treatment are assumed to be non-infectious for the duration of the treatment period, as viral suppression is achieved rapidly following DAA initiation.
- We assume no progression of liver disease occurs during the acute infection phase, as this period is relatively short, whereas meaningful liver deterioration (e.g., fibrosis progression) typically develops over years rather than weeks or months [14].

1.3 Genotype Distribution

- Two genotype groups are modelled: genotype 3 (GT3, proportion ρ) and all other genotypes (proportion $1-\rho$), reflecting the epidemiology of HCV among PWID in Singapore [20]. Genotype distribution is assumed fixed over time.
- GT3 infection is associated with faster liver disease progression. Relative to other genotypes, GT3 confers excess risks of $r_3^{\mathrm{NC}\to\mathrm{CC}}$ and $r_3^{\mathrm{CC}\to\mathrm{DC}}$ for fibrosis progression, and $r_3^{\mathrm{CC}\to\mathrm{HCC}}$ and $r_3^{\mathrm{DC}\to\mathrm{HCC}}$ for hepatocellular carcinoma development [5].
- As genotype is not tracked at the individual level in the model, effective progression rates at each stage are computed as genotype-weighted composites:

$$\tilde{p}_{c,A\to B} = \rho\, r_3^{A\to B}\, p_3^{A\to B} + (1-\rho)\, p_3^{A\to B}$$

  and effective SVR12 rates are similarly weighted:

$$\bar{\alpha}_s = \rho\, \alpha_3^s + (1-\rho)\, \alpha_o^s \quad \text{for stage } s \in \{\mathrm{NC},\ \mathrm{DC}^-,\ \mathrm{DC}^+\}.$$

1.4 Treatment

- A single pan-genotypic DAA regimen, sofosbuvir/velpatasvir (SOF/VEL), is modelled for all treatment scenarios, consistent with AASLD and EASL recommendations [12]. Treatment is administered as one tablet daily for 12 weeks ($\iota_2 = 84$ days).
- The same treatment regimen is applied across all four chronic disease stages (non-cirrhosis, compensated cirrhosis, decompensated cirrhosis, and HCC). Treatment initiation rates τ_k for $k = 1, 2, 3, 4$ are scenario-defined and may differ across stages to reflect access constraints or clinical prioritisation.
- For non-cirrhosis and compensated cirrhosis (stages 1–2), SOF/VEL is administered without ribavirin. For decompensated cirrhosis (stage 3), SOF/VEL is administered with ribavirin (RBV+) where tolerated, yielding SVR12 rates $\alpha_3^{\mathrm{DC}+}$ and $\alpha_o^{\mathrm{DC}+}$, or without ribavirin (RBV−) otherwise, consistent with EASL guidelines [12, 19].
- For HCC (stage 4), no dedicated SOF/VEL SVR12 trial data exist. We conservatively assume the same SVR12 rates as for decompensated cirrhosis, justified by the shared advanced hepatic dysfunction underlying both states and supported by evidence that active HCC is an independent predictor of DAA treatment failure [2].
- A proportion $\bar{\alpha}_s$ of treated individuals achieve SVR12 and transition to the post-SVR susceptible state. The remaining proportion $(1-\bar{\alpha}_s)$ fail treatment and return to the chronic infected state at their respective disease stage.

1.5 Disease Progression

- Chronic HCV infection progresses unidirectionally through four liver disease stages: non-cirrhosis (stage 1), compensated cirrhosis (stage 2), decompensated cirrhosis (stage 3), and hepatocellular carcinoma (stage 4). No regression between stages is permitted in the untreated or treatment-failed states.

- Direct progression from compensated cirrhosis to HCC (stage 2 $\rightarrow$ 4) is permitted in addition to the sequential pathway (2 $\rightarrow$ 3 $\rightarrow$ 4), consistent with observed clinical trajectories [6].
- Following SVR, disease progression is modified as follows, consistent with [10, 17]:
  - For pre-cirrhotic individuals (stage 1), SVR is assumed to halt further progression entirely. No transition 1 $\rightarrow$ 2 occurs in the post-SVR susceptible (u) compartment.
  - For compensated cirrhosis (stage 2), SVR substantially reduces but does not eliminate progression risk. Post-SVR progression rates are $\tilde{p}_{u,2\to3} = \phi^{\mathrm{CC}\to\mathrm{DC}}\,\tilde{p}_{c,2\to3}$ and $\tilde{p}_{u,2\to4} = \phi^{\mathrm{CC}\to\mathrm{HCC}}\,\tilde{p}_{c,2\to4}$.
  - For decompensated cirrhosis (stage 3), SVR does not meaningfully reduce HCC progression, so $\phi^{\mathrm{DC}\to\mathrm{HCC}} = 1.0$ is assumed [10].
- Fibrosis regression following SVR is not modelled. This is a conservative assumption reflecting ongoing uncertainty regarding the extent and clinical significance of fibrosis reversal post-SVR in advanced disease.

1.6 Mortality

- Background (all-cause) mortality μ_i is age-varying and applied to all compartments, fitted to Singapore life-table data (SingStat).
- Disease-specific excess mortality is applied at advanced stages only: μ^{DC} in decompensated cirrhosis (stage 3) and μ^{HCC} in HCC (stage 4), added to background mortality [6].
- SVR reduces disease-specific mortality in stages 3 and 4. For individuals currently on treatment (compartments t), mortality is reduced by factors ψ^{DC} and ψ^{HCC} respectively, yielding $\mu^{\mathrm{DC}}_{\mathrm{SVR}} = \psi^{\mathrm{DC}}\,\mu^{\mathrm{DC}}$ and $\mu^{\mathrm{HCC}}_{\mathrm{SVR}} = \psi^{\mathrm{HCC}}\,\mu^{\mathrm{HCC}}$ [13, 2]. This mortality benefit is applied to the treatment compartment (t) only, as a conservative proxy for post-SVR survival gain, given that SVR status is realised at the end of treatment.
- No mortality benefit from SVR is assumed for stages 1 and 2, consistent with the very low disease-specific mortality at these stages.

2 Considerations

2.1 Treatment Regimen

A single pan-genotypic regimen, sofosbuvir/velpatasvir (SOF/VEL) for 12 weeks, was applied uniformly across all stages of chronic HCV infection. This choice is consistent with the call to action for HCV elimination endorsed by AASLD, EASL, the Asian Pacific Association for the Study of the Liver, and the Latin American Association for the Study of the Liver, presented at the AASLD Liver Meeting in November 2019 [7]. Notably, in a large real-world cohort, all patients with unknown genotype, unknown fibrosis score, and unknown treatment history achieved virological cure with SOF/VEL for 12 weeks, supporting its use in simplified management protocols without the need for extensive pre- or on-treatment monitoring [7].

2.2 Ribavirin Eligibility and Co-administration

Individual ribavirin eligibility was not modelled explicitly, as complete patient-level haematological and physiological data required for eligibility assessment were unavailable. Instead, ribavirin co-administration was incorporated at the population level for the subgroup most likely to benefit clinically.

A recent meta-analysis [19] found no significant difference in SVR12 rates between SOF/VEL with ribavirin for 12 weeks and SOF/VEL without ribavirin for 24 weeks in the general HCV population — contradicting AASLD and EASL guidance recommending the latter for ribavirin-ineligible patients. Furthermore, the addition of ribavirin to SOF/VEL was associated with significantly higher rates of treatment discontinuation, adverse events, serious adverse events, and ribavirin-specific toxicities, including anaemia, fatigue, dyspnoea, and hyperbilirubinaemia [19]. Accordingly, ribavirin was not added to SOF/VEL for the general population.

However, in subgroup analyses, the addition of ribavirin to SOF/VEL significantly improved SVR12 rates specifically in patients with decompensated cirrhosis and genotype 3 infection [19]. Ribavirin co-administration was therefore retained in the model for this subgroup, reflected in the genotype-stratified SVR12 parameters $\alpha_3^{\mathrm{DC}+}$ and $\alpha_o^{\mathrm{DC}+}$.

2.3 SVR12 Rates in Hepatocellular Carcinoma

Current clinical guidelines from EASL and AASLD determine antiviral treatment selection primarily by hepatic functional status — in particular, the presence of decompensated cirrhosis (Child–Pugh B/C) — rather than by HCC status. SOF/VEL-based regimens are recommended for decompensated cirrhosis irrespective of HCC co-occurrence, and no separate antiviral efficacy estimates are provided specifically for patients with concurrent HCC. In the absence of HCC-specific SVR12 data, and given that patients with HCC are managed within the same liver function–based treatment framework as decompensated cirrhosis patients, we assumed equivalent SVR12 rates for HCC and decompensated cirrhosis. This is a conservative assumption; empirical evidence suggests that active HCC is independently associated with reduced DAA treatment response (Section **??**).

2.4 Progression Parameters and Genotype Distribution

Baseline disease progression rates were taken from [6], which was selected on the basis of its similar genotype distribution to that observed in Singapore's HCV population.

Derivation of Genotype-3 Relative Risk

To estimate the relative risk of disease progression for genotype 3 compared with all other genotypes combined, a composite reference hazard ratio was constructed using prevalence-weighted geometric averaging under the proportional hazards assumption. Let HR_g denote the adjusted hazard ratio for genotype g relative to genotype 1, and let p_g denote the prevalence weight of genotype g within the non-genotype-3 population. The composite hazard ratio for other genotypes was:

$$HR_{\text{other}} = \exp\left(\sum_{g\in\{1,2,4\}} p_g \log HR_g\right),$$

and the relative hazard comparing genotype 3 with this composite reference was:

$$HR_{3\,\text{vs other}} = \frac{HR_3}{HR_{\text{other}}}.$$

Uncertainty was propagated via the delta method on the log scale, with standard errors derived from reported 95% confidence intervals as $SE(\log HR) = [\log(\text{Upper}) - \log(\text{Lower})] \,/\, (1.96 \times 2)$. The resulting estimates were:

$$r_3^{\text{NC}\to\text{CC}} = r_3^{\text{CC}\to\text{DC}} = 1.36 \text{ (95\% CI : 1.27–1.46)},$$

$$r_3^{\text{CC}\to\text{HCC}} = r_3^{\text{DC}\to\text{HCC}} = 1.93 \text{ (95\% CI : 1.71–2.17)}.$$

2.5 Post-SVR Progression Rates

The relative risk of progression from compensated cirrhosis to decompensated cirrhosis following SVR ($\phi^{\text{CC}\to\text{DC}}$) was taken from [17], in which liver failure was defined as a composite of ascites, variceal bleeding, jaundice, and overt hepatic encephalopathy — clinical events closely aligned with the definition of decompensated cirrhosis used in this model. This estimate is used as the best available proxy for $\phi^{\text{CC}\to\text{DC}}$ and should be interpreted with caution, as the composite outcome does not map exactly to the model's health state boundary.

3 Model equations

3.1 Effective Rate Definitions

To account for genotype-3 excess risk, define the effective progression rates for each transition as weighted composites across genotypes:

$$\begin{aligned}
\tilde{p}_{c,1\to2} &= \rho\, r_3^{\text{NC}\to\text{CC}}\, p_3^{\text{NC}\to\text{CC}} + (1-\rho)\, p_3^{\text{NC}\to\text{CC}} \\
\tilde{p}_{c,2\to3} &= \rho\, r_3^{\text{CC}\to\text{DC}}\, p_3^{\text{CC}\to\text{DC}} + (1-\rho)\, p_3^{\text{CC}\to\text{DC}} \\
\tilde{p}_{c,2\to4} &= \rho\, r_3^{\text{CC}\to\text{HCC}}\, p_3^{\text{CC}\to\text{HCC}} + (1-\rho)\, p_3^{\text{CC}\to\text{HCC}} \\
\tilde{p}_{c,3\to4} &= \rho\, r_3^{\text{DC}\to\text{HCC}}\, p_3^{\text{DC}\to\text{HCC}} + (1-\rho)\, p_3^{\text{DC}\to\text{HCC}}
\end{aligned}$$

and the post-SVR effective progression rates for recovered individuals:

$$\tilde{p}_{r,2\to3} = \phi^{\mathrm{CC}\to\mathrm{DC}}\,\tilde{p}_{c,2\to3}$$
$$\tilde{p}_{r,2\to4} = \phi^{\mathrm{CC}\to\mathrm{HCC}}\,\tilde{p}_{c,2\to4}$$
$$\tilde{p}_{r,3\to4} = \phi^{\mathrm{DC}\to\mathrm{HCC}}\,\tilde{p}_{c,3\to4}$$

The five HCV states are u (susceptible/uninfected), a (acute infection), c (chronic infection), t (on treatment), and r (recovered). The state transitions are

$$a \xrightarrow{\kappa/\iota_1} r,$$
$$a \xrightarrow{(1-\kappa)/\iota_1} c,$$
$$c \xrightarrow{\tau_k} t,$$
$$t \xrightarrow{\bar{\alpha}_k/\iota_2} r,$$
$$t \xrightarrow{(1-\bar{\alpha}_k)/\iota_2} c,$$
$$r \xrightarrow{\gamma_{i,j}(t)} a.$$

For the active injecting strata D and F, recovered individuals are exposed to reinfection. In accordance with the existing model, J and X do not have a transmission term because they are not active injectors.
The effective SVR12 rate at each stage accounts for the genotype mix:

$$\bar{\alpha}_1 = \rho\,\alpha_3^{\mathrm{NC}} + (1-\rho)\,\alpha_o^{\mathrm{NC}}$$
$$\bar{\alpha}_3^{\pm} = \rho\,\alpha_3^{\mathrm{DC}\pm} + (1-\rho)\,\alpha_o^{\mathrm{DC}\pm}$$

For stages 1–2 (non-cirrhosis and compensated cirrhosis), $\bar{\alpha} \equiv \bar{\alpha}_1$. For stage 3 (decompensated cirrhosis), $\bar{\alpha} \equiv \bar{\alpha}_3^{+}$ (RBV+) or $\bar{\alpha}_3^{-}$ (RBV-free) depending on the treatment scenario. For stage 4 (HCC), we assume $\bar{\alpha} \equiv \bar{\alpha}_3^{+}$ as a conservative approximation.

The effective disease-specific mortality rates incorporating SVR benefit are:

$$\omega\mu_{\mathrm{SVR}}^{\mathrm{DC}} = \psi^{\mathrm{DC}}\,(\omega\mu^{\mathrm{DC}} + \omega\mu_i)$$
$$\omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}} = \psi^{\mathrm{HCC}}\,(\omega\mu^{\mathrm{HCC}} + \omega\mu_i)$$

3.2 Never Incarcerated (D)

$$\frac{dD_{u,1,i}}{dt} = -\big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i\big)\, D_{u,1,i} + \beta_i(t)$$

$$\frac{dD_{a,1,i}}{dt} = \gamma_{i,j}(t)\, D_{u,1,i} + \gamma_{i,j}(t)\, D_{r,1,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(1)} + \omega\mu_i\right) D_{a,1,i}$$

$$\frac{dD_{c,1,i}}{dt} = \frac{1-\kappa}{\iota_1}\, D_{a,1,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\, D_{t,1,i} - \big(\tau_1 + \lambda_i^{(1)} + \omega\mu_i + \tilde{p}_{c,1\to 2}\big)\, D_{c,1,i}$$

$$\frac{dD_{t,1,i}}{dt} = \tau_1\, D_{c,1,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(1)} + \omega\mu_i\right) D_{t,1,i}$$

$$\frac{dD_{r,1,i}}{dt} = \frac{\kappa}{\iota_1}\, D_{a,1,i} + \frac{\bar{\alpha}_1}{\iota_2}\, D_{t,1,i} - \big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i\big)\, D_{r,1,i}$$

$$\frac{dD_{u,2,i}}{dt} = -\big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i\big)\, D_{u,2,i}$$

$$\frac{dD_{a,2,i}}{dt} = \gamma_{i,j}(t)\, D_{u,2,i} + \gamma_{i,j}(t)\, D_{r,2,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(1)} + \omega\mu_i\right) D_{a,2,i}$$

$$\frac{dD_{c,2,i}}{dt} = \frac{1-\kappa}{\iota_1}\, D_{a,2,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\, D_{t,2,i} + \tilde{p}_{c,1\to 2}\, D_{c,1,i} - \big(\tau_2 + \lambda_i^{(1)} + \omega\mu_i + \tilde{p}_{c,2\to 3} + \tilde{p}_{c,2\to 4}\big)\, D_{c,2,i}$$

$$\frac{dD_{t,2,i}}{dt} = \tau_2\, D_{c,2,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(1)} + \omega\mu_i\right) D_{t,2,i}$$

$$\frac{dD_{r,2,i}}{dt} = \frac{\kappa}{\iota_1}\, D_{a,2,i} + \frac{\bar{\alpha}_1}{\iota_2}\, D_{t,2,i} - \big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i + \tilde{p}_{r,2\to 3} + \tilde{p}_{r,2\to 4}\big)\, D_{r,2,i}$$

$$\frac{dD_{u,3,i}}{dt} = -\big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i + \omega\mu^{\mathrm{DC}}\big)\, D_{u,3,i}$$

$$\frac{dD_{a,3,i}}{dt} = \gamma_{i,j}(t)\, D_{u,3,i} + \gamma_{i,j}(t)\, D_{r,3,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(1)} + \omega\mu_i + \omega\mu^{\mathrm{DC}}\right) D_{a,3,i}$$

$$\frac{dD_{c,3,i}}{dt} = \frac{1-\kappa}{\iota_1}\, D_{a,3,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\, D_{t,3,i} + \tilde{p}_{c,2\to 3}\, D_{c,2,i} - \big(\tau_3 + \lambda_i^{(1)} + \omega\mu_i + \omega\mu^{\mathrm{DC}} + \tilde{p}_{c,3\to 4}\big)\, D_{c,3,i}$$

$$\frac{dD_{t,3,i}}{dt} = \tau_3\, D_{c,3,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(1)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}}\right) D_{t,3,i}$$

$$\frac{dD_{r,3,i}}{dt} = \frac{\kappa}{\iota_1}\, D_{a,3,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\, D_{t,3,i} + \tilde{p}_{r,2\to 3}\, D_{r,2,i} - \big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}} + \tilde{p}_{r,3\to 4}\big)\, D_{r,3,i}$$

$$\frac{dD_{u,4,i}}{dt} = -\big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\big)\, D_{u,4,i}$$

$$\frac{dD_{a,4,i}}{dt} = \gamma_{i,j}(t)\, D_{u,4,i} + \gamma_{i,j}(t)\, D_{r,4,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(1)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) D_{a,4,i}$$

$$\frac{dD_{c,4,i}}{dt} = \frac{1-\kappa}{\iota_1}\, D_{a,4,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\, D_{t,4,i} + \tilde{p}_{c,2\to 4}\, D_{c,2,i} + \tilde{p}_{c,3\to 4}\, D_{c,3,i} - \big(\tau_4 + \lambda_i^{(1)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\big)\, D_{c,4,i}$$

$$\frac{dD_{t,4,i}}{dt} = \tau_4\, D_{c,4,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(1)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\right) D_{t,4,i}$$

$$\frac{dD_{r,4,i}}{dt} = \frac{\kappa}{\iota_1}\, D_{a,4,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\, D_{t,4,i} + \tilde{p}_{r,2\to 4}\, D_{r,2,i} + \tilde{p}_{r,3\to 4}\, D_{r,3,i} - \big(\gamma_{i,j}(t) + \lambda_i^{(1)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\big)\, D_{r,4,i}$$

3.3 Currently Incarcerated (J)

$$\frac{dJ_{u,1,i}}{dt} = \lambda_i^{(1)}\, D_{u,1,i} + \lambda_i^{(3)}\, F_{u,1,i} - \left(\lambda_i^{(2)} + \omega\mu_i\right) J_{u,1,i}$$

$$\frac{dJ_{a,1,i}}{dt} = \lambda_i^{(1)}\, D_{a,1,i} + \lambda_i^{(3)}\, F_{a,1,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(2)} + \omega\mu_i\right) J_{a,1,i}$$

$$\frac{dJ_{c,1,i}}{dt} = \lambda_i^{(1)}\, D_{c,1,i} + \lambda_i^{(3)}\, F_{c,1,i} + \frac{1-\kappa}{\iota_1}\, J_{a,1,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\, J_{t,1,i} - \left(\tau_1 + \lambda_i^{(2)} + \omega\mu_i + \tilde{p}_{c,1\to 2}\right) J_{c,1,i}$$

$$\frac{dJ_{t,1,i}}{dt} = \lambda_i^{(1)}\, D_{t,1,i} + \lambda_i^{(3)}\, F_{t,1,i} + \tau_1\, J_{c,1,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(2)} + \omega\mu_i\right) J_{t,1,i}$$

$$\frac{dJ_{r,1,i}}{dt} = \lambda_i^{(1)}\, D_{r,1,i} + \lambda_i^{(3)}\, F_{r,1,i} + \frac{\kappa}{\iota_1}\, J_{a,1,i} + \frac{\bar{\alpha}_1}{\iota_2}\, J_{t,1,i} - \left(\lambda_i^{(2)} + \omega\mu_i\right) J_{r,1,i}$$

$$\frac{dJ_{u,2,i}}{dt} = \lambda_i^{(1)}\, D_{u,2,i} + \lambda_i^{(3)}\, F_{u,2,i} - \left(\lambda_i^{(2)} + \omega\mu_i\right) J_{u,2,i}$$

$$\frac{dJ_{a,2,i}}{dt} = \lambda_i^{(1)}\, D_{a,2,i} + \lambda_i^{(3)}\, F_{a,2,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(2)} + \omega\mu_i\right) J_{a,2,i}$$

$$\frac{dJ_{c,2,i}}{dt} = \lambda_i^{(1)}\, D_{c,2,i} + \lambda_i^{(3)}\, F_{c,2,i} + \frac{1-\kappa}{\iota_1}\, J_{a,2,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\, J_{t,2,i} + \tilde{p}_{c,1\to 2}\, J_{c,1,i} - \left(\tau_2 + \lambda_i^{(2)} + \omega\mu_i + \tilde{p}_{c,2\to 3} + \tilde{p}_{c,2\to 4}\right) J_{c,2,i}$$

$$\frac{dJ_{t,2,i}}{dt} = \lambda_i^{(1)}\, D_{t,2,i} + \lambda_i^{(3)}\, F_{t,2,i} + \tau_2\, J_{c,2,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(2)} + \omega\mu_i\right) J_{t,2,i}$$

$$\frac{dJ_{r,2,i}}{dt} = \lambda_i^{(1)}\, D_{r,2,i} + \lambda_i^{(3)}\, F_{r,2,i} + \frac{\kappa}{\iota_1}\, J_{a,2,i} + \frac{\bar{\alpha}_1}{\iota_2}\, J_{t,2,i} - \left(\lambda_i^{(2)} + \omega\mu_i + \tilde{p}_{r,2\to 3} + \tilde{p}_{r,2\to 4}\right) J_{r,2,i}$$

$$\frac{dJ_{u,3,i}}{dt} = \lambda_i^{(1)}\, D_{u,3,i} + \lambda_i^{(3)}\, F_{u,3,i} - \left(\lambda_i^{(2)} + \omega\mu_i + \omega\mu^{\mathrm{DC}}\right) J_{u,3,i}$$

$$\frac{dJ_{a,3,i}}{dt} = \lambda_i^{(1)}\, D_{a,3,i} + \lambda_i^{(3)}\, F_{a,3,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(2)} + \omega\mu_i + \omega\mu^{\mathrm{DC}}\right) J_{a,3,i}$$

$$\frac{dJ_{c,3,i}}{dt} = \lambda_i^{(1)}\, D_{c,3,i} + \lambda_i^{(3)}\, F_{c,3,i} + \frac{1-\kappa}{\iota_1}\, J_{a,3,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\, J_{t,3,i} + \tilde{p}_{c,2\to 3}\, J_{c,2,i} - \left(\tau_3 + \lambda_i^{(2)} + \omega\mu_i + \omega\mu^{\mathrm{DC}} + \tilde{p}_{c,3\to 4}\right) J_{c,3,i}$$

$$\frac{dJ_{t,3,i}}{dt} = \lambda_i^{(1)}\, D_{t,3,i} + \lambda_i^{(3)}\, F_{t,3,i} + \tau_3\, J_{c,3,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(2)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}}\right) J_{t,3,i}$$

$$\frac{dJ_{r,3,i}}{dt} = \lambda_i^{(1)}\, D_{r,3,i} + \lambda_i^{(3)}\, F_{r,3,i} + \frac{\kappa}{\iota_1}\, J_{a,3,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\, J_{t,3,i} + \tilde{p}_{r,2\to 3}\, J_{r,2,i} - \left(\lambda_i^{(2)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}} + \tilde{p}_{r,3\to 4}\right) J_{r,3,i}$$

$$\frac{dJ_{u,4,i}}{dt} = \lambda_i^{(1)}\, D_{u,4,i} + \lambda_i^{(3)}\, F_{u,4,i} - \left(\lambda_i^{(2)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) J_{u,4,i}$$

$$\frac{dJ_{a,4,i}}{dt} = \lambda_i^{(1)}\, D_{a,4,i} + \lambda_i^{(3)}\, F_{a,4,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(2)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) J_{a,4,i}$$

$$\frac{dJ_{c,4,i}}{dt} = \lambda_i^{(1)}\, D_{c,4,i} + \lambda_i^{(3)}\, F_{c,4,i} + \frac{1-\kappa}{\iota_1}\, J_{a,4,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\, J_{t,4,i} + \tilde{p}_{c,2\to 4}\, J_{c,2,i} + \tilde{p}_{c,3\to 4}\, J_{c,3,i} - \left(\tau_4 + \lambda_i^{(2)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) J_{c,4,}$$

$$\frac{dJ_{t,4,i}}{dt} = \lambda_i^{(1)}\, D_{t,4,i} + \lambda_i^{(3)}\, F_{t,4,i} + \tau_4\, J_{c,4,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(2)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\right) J_{t,4,i}$$

$$\frac{dJ_{r,4,i}}{dt} = \lambda_i^{(1)}\, D_{r,4,i} + \lambda_i^{(3)}\, F_{r,4,i} + \frac{\kappa}{\iota_1}\, J_{a,4,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\, J_{t,4,i} + \tilde{p}_{r,2\to 4}\, J_{r,2,i} + \tilde{p}_{r,3\to 4}\, J_{r,3,i} - \left(\lambda_i^{(2)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\right) J_{r,4,i}$$

3.4 Ever-Incarcerated (F)

$$\frac{dF_{u,1,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{u,1,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i\right) F_{u,1,i}$$

$$\frac{dF_{a,1,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{a,1,i} + \gamma_{i,j}(t)\,F_{u,1,i} + \gamma_{i,j}(t)\,F_{r,1,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(3)} + \omega\mu_i\right) F_{a,1,i}$$

$$\frac{dF_{c,1,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{c,1,i} + \frac{1-\kappa}{\iota_1}\,F_{a,1,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\,F_{t,1,i} - \left(\tau_1 + \lambda_i^{(3)} + \omega\mu_i + \tilde{p}_{c,1\to 2}\right) F_{c,1,i}$$

$$\frac{dF_{t,1,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{t,1,i} + \tau_1\,F_{c,1,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(3)} + \omega\mu_i\right) F_{t,1,i}$$

$$\frac{dF_{r,1,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{r,1,i} + \frac{\kappa}{\iota_1}\,F_{a,1,i} + \frac{\bar{\alpha}_1}{\iota_2}\,F_{t,1,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i\right) F_{r,1,i}$$

$$\frac{dF_{u,2,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{u,2,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i\right) F_{u,2,i}$$

$$\frac{dF_{a,2,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{a,2,i} + \gamma_{i,j}(t)\,F_{u,2,i} + \gamma_{i,j}(t)\,F_{r,2,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(3)} + \omega\mu_i\right) F_{a,2,i}$$

$$\frac{dF_{c,2,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{c,2,i} + \frac{1-\kappa}{\iota_1}\,F_{a,2,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\,F_{t,2,i} + \tilde{p}_{c,1\to 2}\,F_{c,1,i} - \left(\tau_2 + \lambda_i^{(3)} + \omega\mu_i + \tilde{p}_{c,2\to 3} + \tilde{p}_{c,2\to 4}\right) F_{c,2,i}$$

$$\frac{dF_{t,2,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{t,2,i} + \tau_2\,F_{c,2,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(3)} + \omega\mu_i\right) F_{t,2,i}$$

$$\frac{dF_{r,2,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{r,2,i} + \frac{\kappa}{\iota_1}\,F_{a,2,i} + \frac{\bar{\alpha}_1}{\iota_2}\,F_{t,2,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i + \tilde{p}_{r,2\to 3} + \tilde{p}_{r,2\to 4}\right) F_{r,2,i}$$

$$\frac{dF_{u,3,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{u,3,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i + \omega\mu^{\mathrm{DC}}\right) F_{u,3,i}$$

$$\frac{dF_{a,3,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{a,3,i} + \gamma_{i,j}(t)\,F_{u,3,i} + \gamma_{i,j}(t)\,F_{r,3,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(3)} + \omega\mu_i + \omega\mu^{\mathrm{DC}}\right) F_{a,3,i}$$

$$\frac{dF_{c,3,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{c,3,i} + \frac{1-\kappa}{\iota_1}\,F_{a,3,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\,F_{t,3,i} + \tilde{p}_{c,2\to 3}\,F_{c,2,i} - \left(\tau_3 + \lambda_i^{(3)} + \omega\mu_i + \omega\mu^{\mathrm{DC}} + \tilde{p}_{c,3\to 4}\right) F_{c,3,i}$$

$$\frac{dF_{t,3,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{t,3,i} + \tau_3\,F_{c,3,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(3)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}}\right) F_{t,3,i}$$

$$\frac{dF_{r,3,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{r,3,i} + \frac{\kappa}{\iota_1}\,F_{a,3,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\,F_{t,3,i} + \tilde{p}_{r,2\to 3}\,F_{r,2,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}} + \tilde{p}_{r,3\to 4}\right) F_{r,3,i}$$

$$\frac{dF_{u,4,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{u,4,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) F_{u,4,i}$$

$$\frac{dF_{a,4,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{a,4,i} + \gamma_{i,j}(t)\,F_{u,4,i} + \gamma_{i,j}(t)\,F_{r,4,i} - \left(\frac{1}{\iota_1} + \lambda_i^{(3)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) F_{a,4,i}$$

$$\frac{dF_{c,4,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{c,4,i} + \frac{1-\kappa}{\iota_1}\,F_{a,4,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\,F_{t,4,i} + \tilde{p}_{c,2\to 4}\,F_{c,2,i} + \tilde{p}_{c,3\to 4}\,F_{c,3,i} - \left(\tau_4 + \lambda_i^{(3)} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) F_{c,4,i}$$

$$\frac{dF_{t,4,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{t,4,i} + \tau_4\,F_{c,4,i} - \left(\frac{1}{\iota_2} + \lambda_i^{(3)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\right) F_{t,4,i}$$

$$\frac{dF_{r,4,i}}{dt} = \pi\,\lambda_i^{(2)}\,J_{r,4,i} + \frac{\kappa}{\iota_1}\,F_{a,4,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\,F_{t,4,i} + \tilde{p}_{r,2\to 4}\,F_{r,2,i} + \tilde{p}_{r,3\to 4}\,F_{r,3,i} - \left(\gamma_{i,j}(t) + \lambda_i^{(3)} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\right) F_{r,4,i}$$

3.5 Former PWID (X)

$$\frac{dX_{u,1,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{u,1,i} - \omega\mu_i\,X_{u,1,i}$$

$$\frac{dX_{a,1,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{a,1,i} - \left(\frac{1}{\iota_1} + \omega\mu_i\right) X_{a,1,i}$$

$$\frac{dX_{c,1,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{c,1,i} + \frac{1-\kappa}{\iota_1}\,X_{a,1,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\,X_{t,1,i} - \left(\tau_1 + \omega\mu_i + \tilde{p}_{c,1\to 2}\right) X_{c,1,i}$$

$$\frac{dX_{t,1,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{t,1,i} + \tau_1\,X_{c,1,i} - \left(\frac{1}{\iota_2} + \omega\mu_i\right) X_{t,1,i}$$

$$\frac{dX_{r,1,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{r,1,i} + \frac{\kappa}{\iota_1}\,X_{a,1,i} + \frac{\bar{\alpha}_1}{\iota_2}\,X_{t,1,i} - \omega\mu_i\,X_{r,1,i}$$

$$\frac{dX_{u,2,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{u,2,i} - \omega\mu_i\,X_{u,2,i}$$

$$\frac{dX_{a,2,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{a,2,i} - \left(\frac{1}{\iota_1} + \omega\mu_i\right) X_{a,2,i}$$

$$\frac{dX_{c,2,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{c,2,i} + \frac{1-\kappa}{\iota_1}\,X_{a,2,i} + \frac{1-\bar{\alpha}_1}{\iota_2}\,X_{t,2,i} + \tilde{p}_{c,1\to 2}\,X_{c,1,i} - \left(\tau_2 + \omega\mu_i + \tilde{p}_{c,2\to 3} + \tilde{p}_{c,2\to 4}\right) X_{c,2,i}$$

$$\frac{dX_{t,2,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{t,2,i} + \tau_2\,X_{c,2,i} - \left(\frac{1}{\iota_2} + \omega\mu_i\right) X_{t,2,i}$$

$$\frac{dX_{r,2,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{r,2,i} + \frac{\kappa}{\iota_1}\,X_{a,2,i} + \frac{\bar{\alpha}_1}{\iota_2}\,X_{t,2,i} - \left(\omega\mu_i + \tilde{p}_{r,2\to 3} + \tilde{p}_{r,2\to 4}\right) X_{r,2,i}$$

$$\frac{dX_{u,3,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{u,3,i} - \left(\omega\mu_i + \omega\mu^{\mathrm{DC}}\right) X_{u,3,i}$$

$$\frac{dX_{a,3,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{a,3,i} - \left(\frac{1}{\iota_1} + \omega\mu_i + \omega\mu^{\mathrm{DC}}\right) X_{a,3,i}$$

$$\frac{dX_{c,3,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{c,3,i} + \frac{1-\kappa}{\iota_1}\,X_{a,3,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\,X_{t,3,i} + \tilde{p}_{c,2\to 3}\,X_{c,2,i} - \left(\tau_3 + \omega\mu_i + \omega\mu^{\mathrm{DC}} + \tilde{p}_{c,3\to 4}\right) X_{c,3,i}$$

$$\frac{dX_{t,3,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{t,3,i} + \tau_3\,X_{c,3,i} - \left(\frac{1}{\iota_2} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}}\right) X_{t,3,i}$$

$$\frac{dX_{r,3,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{r,3,i} + \frac{\kappa}{\iota_1}\,X_{a,3,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\,X_{t,3,i} + \tilde{p}_{r,2\to 3}\,X_{r,2,i} - \left(\omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{DC}} + \tilde{p}_{r,3\to 4}\right) X_{r,3,i}$$

$$\frac{dX_{u,4,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{u,4,i} - \left(\omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) X_{u,4,i}$$

$$\frac{dX_{a,4,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{a,4,i} - \left(\frac{1}{\iota_1} + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) X_{a,4,i}$$

$$\frac{dX_{c,4,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{c,4,i} + \frac{1-\kappa}{\iota_1}\,X_{a,4,i} + \frac{1-\bar{\alpha}_3^{\pm}}{\iota_2}\,X_{t,4,i} + \tilde{p}_{c,2\to 4}\,X_{c,2,i} + \tilde{p}_{c,3\to 4}\,X_{c,3,i} - \left(\tau_4 + \omega\mu_i + \omega\mu^{\mathrm{HCC}}\right) X_{c,4,i}$$

$$\frac{dX_{t,4,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{t,4,i} + \tau_4\,X_{c,4,i} - \left(\frac{1}{\iota_2} + \omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\right) X_{t,4,i}$$

$$\frac{dX_{r,4,i}}{dt} = (1-\pi)\,\lambda_i^{(2)}\,J_{r,4,i} + \frac{\kappa}{\iota_1}\,X_{a,4,i} + \frac{\bar{\alpha}_3^{\pm}}{\iota_2}\,X_{t,4,i} + \tilde{p}_{r,2\to 4}\,X_{r,2,i} + \tilde{p}_{r,3\to 4}\,X_{r,3,i} - \left(\omega\mu_i + \omega\mu_{\mathrm{SVR}}^{\mathrm{HCC}}\right) X_{r,4,i}$$

4 Notation

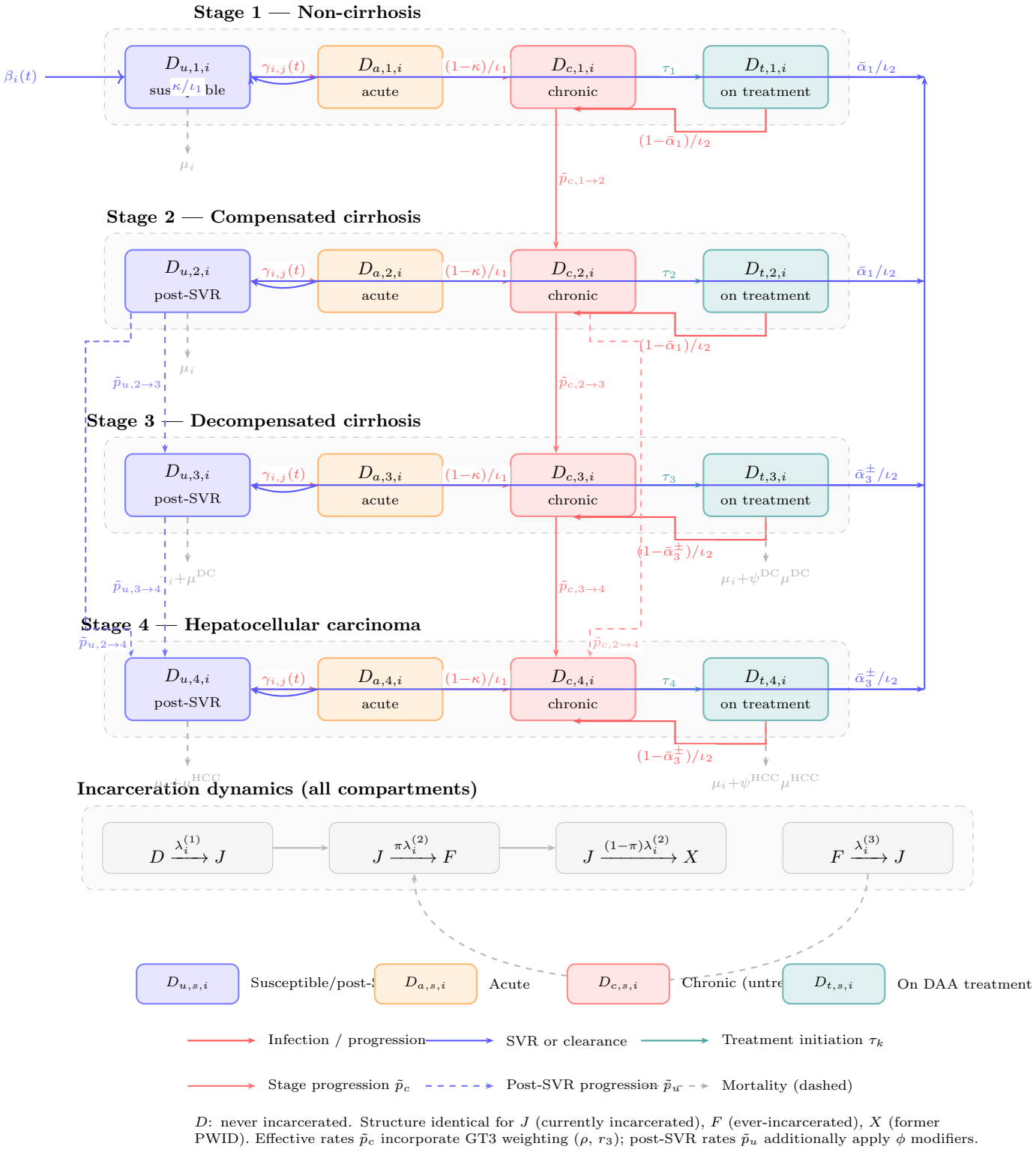
Table 1: Epidemiological and treatment parameters

Parameter	Interpretation	Value	Source
Transmission and natural history			
HCV infection states: For each population stratum, liver-disease stage, and age group, u denotes susceptible/uninfected, a acute infection, c chronic infection, t currently receiving treatment, and r recovered after spontaneous clearance or successful treatment (SVR12). Recovered individuals can be reinfected and return from r to a.			
$\gamma_{i,j}(t)$	Force of HCV infection for age group i from infectious age group j at time t	Model-derived	Assumed from contact and infectious compartments
q	Transmission probability per needle-sharing event	0.0057 $\mathcal{U}(0.0032–0.0105)$	[1]
κ	Probability of spontaneous clearance after acute infection; clearance is $a \to r$	0.26 $\mathcal{U}(0.22–0.29)$	[9]
ι_1	Mean duration of acute infection	180 days	[3]
ι_2	Mean duration of DAA treatment	84 days	[11]
Genotype distribution			
ρ	Proportion with genotype 3 (GT3)	0.78 $\mathcal{U}(0.74–0.82)$	[20]
SVR12 rates by disease stage, genotype, and ribavirin use			
α_3^{NC}	SVR12 rate (GT3; non-cirrhosis & compensated cirrhosis; RBV-free)	0.976 $\mathcal{U}(0.968–0.982)$	[12]
α_o^{NC}	SVR12 rate (other GT; non-cirrhosis & compensated cirrhosis; RBV-free)	0.988 $\mathcal{U}(0.984–0.991)$	[12]
$\alpha_3^{\mathrm{DC-}}$	SVR12 rate (GT3; decompensated cirrhosis; RBV-free)	0.45 $\mathcal{U}(0.13–0.79)$	[12, 19]
$\alpha_o^{\mathrm{DC-}}$	SVR12 rate (other GT; decompensated cirrhosis; RBV-free)	0.90 $\mathcal{U}(0.82–0.96)$	[12, 19]
$\alpha_3^{\mathrm{DC+}}$	SVR12 rate (GT3; decompensated cirrhosis; RBV+)	0.87 $\mathcal{U}(0.71–0.98)$	[12, 19]
$\alpha_o^{\mathrm{DC+}}$	SVR12 rate (other GT; decompensated cirrhosis; RBV+)	0.92 $\mathcal{U}(0.87–0.95)$	[12, 19]
Treatment			
τ_k	Treatment initiation rate under scenario $k = 1, 2, 3, 4$	Scenario-defined	Assumed
Incarceration dynamics			
$\lambda_i^{(1)}$	First-arrest rate for age subgroup $i = 1, \ldots, 9$	Age-varying	Fitted to SPS data
$\lambda_i^{(2)}$	Release rate for age subgroup i	Age-varying	Fitted to SPS data
$\lambda_i^{(3)}$	Re-arrest rate for age subgroup i	Age-varying	Fitted to SPS data
π	Recidivism probability	0.65 $\mathcal{U}(0.54–0.70)$	Fitted to SPS data
Population entry			
$\beta_i(t)$	Entry rate into the PWID population for subgroup i at time t	Time- & age-varying	
N_0	Initial population for all ever-PWIDs in 2017	11000 (95%CI:6500–17000)	[15]
-	HCV seroprevalence among ever-PWIDs in 2017 in Singapore(%)	46.5% Bernoulli($p = 0.465$)	[4, 18]

Table 2: Disease progression and mortality parameters

Parameter	Interpretation	Value	Source
Progression rates (untreated, other genotype baseline)			
$p_3^{\mathrm{NC}\to\mathrm{CC}}$	Annual progression rate: non-cirrhosis → compensated cirrhosis (other GT)	0.027 $\mathcal{N}(0.027, 0.0008^2)$	[16]
$p_3^{\mathrm{CC}\to\mathrm{DC}}$	Annual progression rate: compensated → decompensated cirrhosis (other GT)	0.039 Beta(14.6, 360.2)	[6]
$p_3^{\mathrm{CC}\to\mathrm{HCC}}$	Annual progression rate: compensated cirrhosis → HCC (other GT)	0.014 Beta(1.9, 136.1)	[6]
$p_3^{\mathrm{DC}\to\mathrm{HCC}}$	Annual progression rate: decompensated cirrhosis → HCC (other GT)	0.014 Beta(1.9, 136.1)	[6]
Genotype-3 relative risks for progression			
$r_3^{\mathrm{NC}\to\mathrm{CC}}$	Relative risk of progression: non-cirrhosis → compensated cirrhosis (GT3 vs. other)	1.36 $\mathcal{U}(1.27, 1.46)$	[5]
$r_3^{\mathrm{CC}\to\mathrm{DC}}$	Relative risk of progression: compensated → decompensated cirrhosis (GT3 vs. other)	1.36 $\mathcal{U}(1.27, 1.46)$	[5]
$r_3^{\mathrm{CC}\to\mathrm{HCC}}$	Relative risk of progression: compensated cirrhosis → HCC (GT3 vs. other)	1.93 $\mathcal{U}(1.71, 2.17)$	[5]
$r_3^{\mathrm{DC}\to\mathrm{HCC}}$	Relative risk of progression: decompensated cirrhosis → HCC (GT3 vs. other)	1.93 $\mathcal{U}(1.71, 2.17)$	[5]
Post-SVR relative risks for progression			
$\phi^{\mathrm{CC}\to\mathrm{DC}}$	Relative risk of progression: compensated → decompensated cirrhosis after SVR	0.07 LogN(95% CI: 0.03–0.20)	[17]
$\phi^{\mathrm{CC}\to\mathrm{HCC}}$	Relative risk of progression: compensated cirrhosis → HCC after SVR	0.23 LogN(95% CI: 0.16–0.35)	[10]
$\phi^{\mathrm{DC}\to\mathrm{HCC}}$	Relative risk of progression: decompensated cirrhosis → HCC after SVR	1.00	Assumed
Mortality			
μ_i	Background (all-cause) mortality rate for age subgroup i	Age-varying	Fitted to SingStat data
ω	Standadised Mortality Rate of ever-PWIDS relative to normal people	14.68 (95% CI: 13.01–16.35)	[8]
μ^{DC}	Additional disease mortality rate in decompensated cirrhosis	0.13 Beta(147.0, 984.0)	[6]
μ^{HCC}	Additional disease mortality rate in HCC	0.43 Beta(117.1, 155.2)	[6]
ψ^{DC}	Relative mortality risk in decompensated cirrhosis after SVR	0.45 $\mathcal{U}(0.31, 0.63)$	[13]
ψ^{HCC}	Relative mortality risk in HCC after SVR	0.37 (95% CI: 0.16–0.83)	[2]

5 Model schematic



D: never incarcerated. Structure identical for J (currently incarcerated), F (ever-incarcerated), X (former PWID). Effective rates $\tilde{p}_c$ incorporate GT3 weighting (ρ, r_3); post-SVR rates $\tilde{p}_u$ additionally apply ϕ modifiers.

6 Model calibration

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